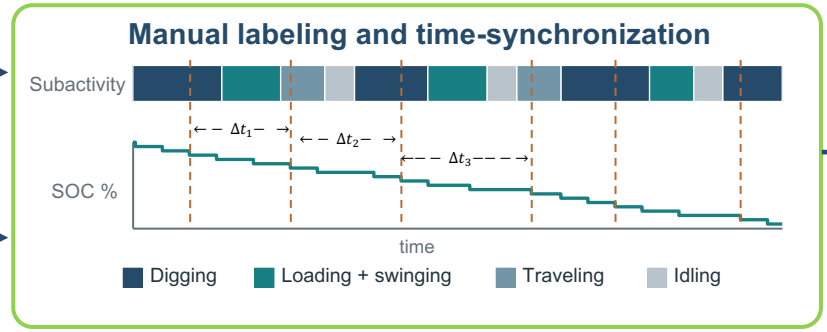
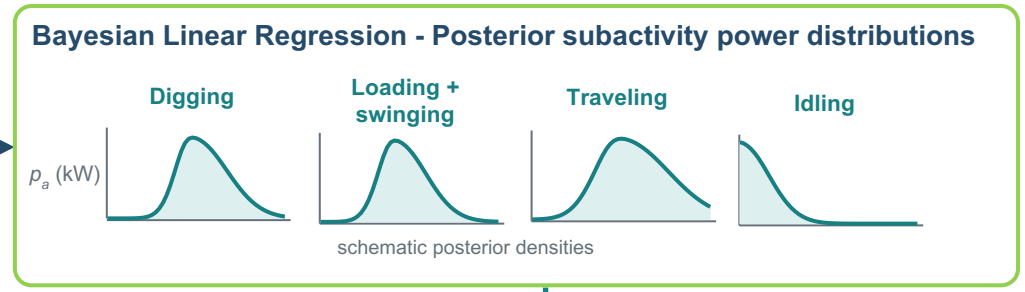
(a) Probabilistic Subactivity Power Estimation

Field videos

SOC telematics





(b) Shrinking-horizon MPC

Stochastic CEV–MCS Simulation Plant



First control input of the Prediction horizon

Updated state

Joint CEV work-charging and MCS scheduling

minimize Energy cost + carbon emissions cost + demand charges + missed-work penalty + MCS towing labor

subject to

1. Charging and discharging power limits of the MCSs and CEVs.
2. Energy consumption and SOE dynamics of the MCSs and CEVs.
3. MCS spatial routing and CEV–MCS connection status.
4. CEV work scheduling (one subactivity per time-step, work durations) .